

HOUSE PRICE PREDICTION

A Machine Learning Regression System for Real Estate Valuation

Python • Pandas • Scikit-learn • Matplotlib



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The Problem & Purpose



Problem Statement

Manual property appraisal is subjective, slow, and prone to human bias. There is no standardized, data-driven way to estimate a fair price.

Purpose

Automate house valuation using historical data and predictive modeling for transparent, objective price estimates.

Objectives

- Train an ML model on historical real estate data
- Evaluate model performance and statistical fit
- Accept dynamic user input (area, bedrooms) for real-time predictions
- Visualize actual vs. predicted price variance



Proposed System & Workflow

An automated pipeline: the user enters house features and the system instantly returns a predicted price.

1

Data Loading

Historical housing data loaded into a Pandas DataFrame

2

Train / Test Split

Dataset divided into training and testing subsets

3

Model Training

Algorithm learns the relationship between features and price

4

Evaluate & Visualize

Scatter plot compares actual vs. predicted prices

5

User Prediction

Console input processed and predicted price displayed



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Dataset & Technologies

D Dataset

Features	Area_sqft, Bedrooms
Target	Price
Data type	Numerical (continuous)
Split	Train / test subsets

T Technology Stack

- **Python** Core programming language
- **Pandas** Data manipulation & DataFrames
- **Scikit-learn** Regression model training & scoring
- **Matplotlib** Actual vs. predicted visualization



Model & Evaluation

R² SCORE

99.74%

The model's inputs (area, bedrooms) explain nearly all of the variance in house price on this dataset.

Note: .score() returns R² for regression — not classification "accuracy."

Algorithm

Regression model — predicts continuous price values

Training

Fit on Area_sqft and Bedrooms against historical Price

Testing

`model.predict(X_test)` generates estimates on unseen data

Visualization

Actual (blue dots) vs. predicted (red dashed line) scatter plot



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Strengths & Limitations

Advantages

- Instantaneous results
- Removes human bias from valuation
- Simple, minimal input required
- Achieves a 99.74% R^2 score

Limitations

- Ignores location, property age, amenities
- Relies on static historical data
- Cannot adapt to sudden market shifts or crashes



Applications & Future Scope

Real-World Applications

- Estimation tool for real estate platforms (e.g., MagicBricks, Zillow)
- Helps investors spot over- or under-valued properties

Future Scope

- Add location via One-Hot Encoding
- Deploy as a web app (Flask / Django / Streamlit)
- Expand features: distance to hospitals, parking, etc.

Conclusion

This project shows that regression-based machine learning can effectively predict property prices from physical attributes. A 99.74% R^2 score confirms the model's strong grasp of the underlying data patterns — a solid foundation for a production-ready valuation tool.

